

Requirements Engineering & Quality Assurance

Assignment 1: Requirements Elicitation via Role-Play

Group of 2 Members | Total: 20 marks | First assignment of the course

Overview

This assignment introduces **requirements elicitation**: the process of drawing out a system's requirements by engaging directly with its stakeholders. Working in pairs, you will role-play the stakeholders of a small system, evaluate the quality of the requirements you elicit against six standard quality attributes (**conflict, ambiguity, completeness, consistency, testability, and evolvability**), and then compare your results against those produced by an AI assistant given only a high-level brief. The final requirements list you produce here is not a one-off exercise, you will reuse it in Assignment 2, where you check it against the behaviour of a running application, so treat it as a deliverable you want to use in a few weeks.

Project Selection

Choose **one** of three projects in different domains:

1. **Library Management**
2. **Inventory Management** and
3. **Equipment Rental Management.**

Rules for how groups select a project will be announced separately, before the assignment opens, to avoid every group converging on the same domain.

Once your group selects a domain, you **remain on it for every assignment this term**. Make sure you use the documents corresponding to the project assigned to your group. All documents for Assignment 1 are provided through OnQ. These include:

- Scenarios for the three domains
- Three stakeholder role cards for your assigned project
- Pre-work checklist and Requirements quality-check worksheet

Important Constraint

Read only your own domain's documents. Your requirements must come entirely from your own domain's role-play, not from any other domain's materials.

Steps

1. **Pre-work (individual, before the group session).** Complete **prework_checklist**. This includes setting up a code editor, Git, a GitHub account, repository creation, and TA invited as a collaborator.
2. **Read the scenario brief as a group.** Review your chosen domain's **scenario_brief** together before role assignment.
3. **Role assignment.** There are 3 stakeholder role cards for each project but only 2 group members. Each member takes one role card as their **primary role**. The third role card is **shared** for both members read and role-play it jointly rather than assigning it to one person.
4. **Individual elicitation.**
 - For your primary role: your partner interviews you in-role, as a requirements analyst would interview a real stakeholder, then you swap, and you interview your partner in their primary role. Each of you writes your **own** individual requirements list from your own primary role, in your **own words**, using standard “the system shall...” requirement statements.
 - For the shared third role: read the role card together, discuss what that stakeholder would realistically want, and write **one joint list** for that role. If the two of you arrive at different interpretations of what that stakeholder wants, do not hide the **disagreement**. Capture it as part of the elicitation process.
 - By the end of this step you should have **3 requirements lists total**: 2 individual list (one per member) + 1 joint list.
5. **Group consolidation.** As a group, merge the two individual and one joint lists into one consolidated requirements list. Run **every** requirement through the quality-check worksheet (conflict, ambiguity, completeness, consistency, testability, evolvability). Pay particular attention to any point where two stakeholders' role cards pulled in different directions. This represents is a real conflict, not an error, and it should be **documented, not silently dropped**.
6. **AI comparison.** Give an AI assistant (e.g., ChatGPT, Gemini) **only** the **scenario brief** not your role cards or your own requirements list. Ask AI assistant to generate a requirements list for the system. Run the same quality-check worksheet on the AI-generated list.
7. **Comparison write-up (0.5–1 page).** Compare your group's consolidated requirements with the AI-generated requirements. Discuss:
 - What did the AI's list miss that your role-play surfaced (especially any stakeholder conflict)?
 - What did the AI invent or assume that wasn't grounded in any actual stakeholder need?
 - Where did the two requirements lists agree?
 - Did the AI identify any useful requirements that your group had not initially considered?

Due date: Sept. 25, 2026 (plus a 3-days automatic extension)

8. **Final requirements list.** Produce your group's final, refined requirements list, incorporating anything useful from the AI comparison, where appropriate while ensuring that the final requirements remain grounded in the project and stakeholder needs. Keep this list as you will compare it against the actual running application's behaviour in Assignment 2.

Deliverables

Submit to OnQ as one PDF; GitHub and pre-work confirmation should be included as specified below.

#	Deliverable
1	Role-based requirements lists: 2 individual (one per member, labeled by role) + 1 joint (shared third role)
2	Group consolidated requirements list + completed quality-check worksheet
3	AI-generated requirements list + completed quality-check worksheet
4	Comparison write-up (own vs. AI) in 0.5 to 1 page comparing your group's requirements with the AI-generated requirements
5	Group's final refined requirements list
6	Pre-work checklist confirmation (repo link, TA invited)

Grading Scheme

Out of **20 marks** for this offering. *Note: this assignment will later be rescaled to a /10 weight in the final grade book.*

Component	Marks (/20)
Role-based requirements lists (2 individual + 1 joint)	3
Group consolidated list + quality-check worksheet	6
AI-generated list + quality-check worksheet	3
Comparison write-up	4
Final requirements list quality	3
Pre-work checklist completion	1
Total	20